

IVY MATOBORI

INSTRUMENTATION & CONTROL ENGINEER

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EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)
Bachelor of Science in Control and Instrumentation Engineering

Nairobi, Kenya
2019 – 2024

TECHNICAL SKILLS

- **Instrumentation & Control:** Sensor and actuator integration, calibration procedures, flow and temperature measurement, data acquisition, real-time process data logging, energy metering, closed-loop control
- **Industrial Automation:** PLC programming, ladder logic, Siemens TIA Portal, HMI configuration, machine control, alarm handling, safety interlocks
- **Industrial Communication:** Modbus, RS485, CAN bus, one-wire, UART, SPI, I²C, LoRa, BLE, NFC, WiFi, 4G/LTE
- **Embedded Systems & Firmware:** Embedded C/C++, FreeRTOS, ESP-IDF, ESP32, bare-metal development, peripheral drivers, MCU selection, low-power design, Embedded Linux (in progress)
- **Electronics & PCB Design:** Altium Designer, KiCad, Fusion 360, schematic capture, multi-layer PCB layout, power architecture and regulation, component selection, DFM, PCBA, board bring-up
- **Debugging & Validation:** Oscilloscope, logic analyzer, multimeter, signal integrity analysis, root cause analysis, system-level testing, design reviews, Git, technical documentation

WORK EXPERIENCE

Quepay Limited, Kenya | *C++ · FreeRTOS · ESP-IDF · Altium · LTE/BLE/I²C*
Hardware Designer & Technical Operations Lead

Feb 2025 – Present
On-site, Nairobi, Kenya

- Sustained a **99%** device uptime SLA across **KES 10M+** in processed transactions, as measured by fleet monitoring, by building the C++ and FreeRTOS firmware that actuates volume-metered dispensing and streams telemetry over LTE.
- Consolidated **6+** machine verticals onto one controller board, from water and milk ATMs to vending and laundry, by designing the field-deployed hardware and **4-layer** PCB in Altium Designer, scaling deployment **4×** in **4 months**.
- Specified, procured, installed, and commissioned controller and sensing hardware across **6** field stations, as measured by site acceptance sign-off, validating each installation on site before hand-off to the operator.
- Drove long-term measurement accuracy on milk ATM dispensers, holding dispensed volume within **±2%** of setpoint, by developing **calibration procedures** for flow and temperature instrumentation and troubleshooting instrument failures in the field.
- Held dispense faults under **1%** across **250** deployed units, by building fault detection, alarm handling, and safety interlocks into the controller firmware, including over-dispense cutoff and automatic payment reversal on incomplete dispense.
- Reduced field return rate **60%** across **250** installed units over **12 months**, by performing board bring-up and signal integrity validation on every hardware revision and driving root cause analysis on **15** production failures.
- Commissioned standardised production test procedures that cut per-unit validation time to **12 minutes** across **40** builds per month, leading a **4-person** team across assembly, inspection, and field deployment.
- Cut board-level defect rate to **2%** across **300+** units shipped, by introducing first-article inspection, incoming component verification, and assembly review into the PCBA workflow.

Veno Autobotics, Kenya | *Level & Energy Metering · PCB Design · LoRa/BLE/NFC*
Embedded Systems Engineer, Embedded Hardware Design

Apr 2024 – Jan 2025
On-site, Nairobi, Kenya

- Shipped **7** IoT products across school, utility, retail, and municipal deployments, by owning embedded hardware design on a **5-engineer** studio team spanning level sensing, energy metering, amusement metering, and public display systems.
- Instrumented liquid level and energy consumption for field deployments, by designing the sensing hardware behind Veno's Level Sensor and VeNode energy meter, scaling metering from a single appliance to a full industrial complex.
- Extended device reach beyond cellular coverage across **4** radio technologies, LoRa, BLE, NFC, and WiFi, by designing low-power RF hardware and selecting the radio per deployment against range, power budget, and site constraints.
- Laid out the smallest PCB in company history at **15 mm** by **20 mm**, by taking client hardware from schematic and layout through prototype assembly to enclosure and packaging design with mechanical engineers.
- Held control loop jitter under **2 ms** on **32 KB** of flash, by developing bare-metal C/C++ firmware implementing **8** peripheral drivers over I²C, SPI, and UART and running end-to-end system verification before design freeze.

Kenya Space Agency, Kenya | *Power Electronics · PCB Design · Systems Engineering*
Satellite Systems Engineer, Electrical Power System (EPS)

Jan 2023 – Mar 2023
On-site, Nairobi, Kenya

- Met a **5 W** orbit-average power budget at **92%** conversion efficiency, as measured by bench characterisation, by architecting the EPS power distribution network around **4** regulated rails on redundant distribution paths.
- Took the **6-layer** power distribution board from schematic capture through layout to fabrication release, integrating DC-DC conversion, filtering, and overcurrent protection for all **4** rails within a **3-month** placement.
- Contained single-point failures to a single branch across **6** independently protected distribution paths, by engineering a fault-tolerant power architecture and selecting overcurrent protection and isolation devices for each branch.
- Cut standby current draw **18 mA** against the baseline design, as measured on the bench, by leading the EPS microcontroller trade study across **6** candidate parts scored on power, peripherals, reliability, and radiation tolerance.
- Carried the EPS design through **2** multidisciplinary design reviews with engineers from **5** national space agencies across Kenya, Uganda, Egypt, Nigeria, and Rwanda, by producing **40+** pages of electrical interface, verification, and test documentation.

- Prototyped **3** microcontroller-based sensor and actuator test rigs in **16 weeks**, validating each against **10** functional acceptance criteria before hand-off to the project team.
- Turned **4** boards from schematic to tested hardware on a **3-week** cycle in KiCad, brought **6** sensor interfaces online in embedded C, and cut prototype idle power draw **35%**.

Atmega328P Development Board | Altium Designer · Open Source

2024

- Replicated the Arduino Uno platform to **50%** on a **40 mm × 50 mm** board, by laying out a standalone Atmega328P in Altium Designer with a CP2102 USB-to-UART bridge replacing the Atmega8, published open source for community fork.